ParkVision AI: Real-Time Smart Parking Space Detection Using YOLOv8

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***Abstract*—Efficient parking management has become a critical challenge in urban environments due to the rapid increase in vehicle density and limited availability of parking infrastructure. Conventional parking systems rely heavily on manual supervision or sensor-based solutions, which are often inefficient, costly, and lack scalability. This paper presents *ParkVision AI*, a real-time smart parking detection system based on deep learning and computer vision techniques. The proposed system utilizes the YOLOv8 object detection model trained on the PKLot dataset to accurately detect and classify parking spaces as either occupied or empty from CCTV or video inputs. An end-to-end pipeline is developed, incorporating object detection, confidence calibration, agnostic non-maximum suppression, and parking space analysis to ensure reliable performance in high-density and complex environments. The system further integrates a Streamlit-based interactive dashboard that provides real-time analytics, including total parking spaces, available spaces, occupied spaces, and occupancy percentage. Experimental evaluation demonstrates that the proposed approach achieves high detection accuracy and is capable of processing large-scale parking scenarios efficiently. The vision-based design eliminates the need for additional hardware, making the system cost-effective and scalable. The proposed solution contributes toward intelligent transportation systems by reducing traffic congestion, optimizing parking utilization, and enabling smart city applications.**

***Index Terms*—YOLOv8, Smart Parking, Computer Vision, Object Detection, PKLot Dataset, Streamlit, Deep Learning**

I. Introduction

Rapid urbanization and the exponential growth in vehicle ownership have made efficient parking management a critical challenge in modern cities. The increasing demand for parking spaces, coupled with limited infrastructure, often leads to traffic congestion, increased fuel consumption, and driver frustration. Conventional parking systems rely heavily on manual supervision or basic sensor-based mechanisms, which are either inefficient, expensive to deploy at scale, or lack real-time adaptability. In many public and private parking facilities such as shopping malls, corporate campuses, and educational institutions, the absence of intelligent monitoring systems results in poor space utilization and increased search time for drivers. Manual monitoring by security personnel is not only labor-intensive but also prone to human error, especially in high-density parking environments. Furthermore, traditional systems fail to provide real-time insights into parking availability, making them unsuitable for smart city applications. To address these challenges, recent advancements in computer vision and deep learning have enabled the development of automated parking management systems. Among these, object detection models such as YOLO (You Only Look Once) have gained significant attention due to their ability to perform real-time detection with high accuracy. These models can process video feeds from surveillance cameras and identify objects efficiently, making them ideal for parking space analysis. In this paper, we present **ParkVision AI**, a real-time smart parking space detection system that leverages the capabilities of the YOLOv8 deep learning model. The proposed system analyzes CCTV or video input streams to detect and classify parking spaces as either *occupied* or *empty*. Unlike traditional approaches, our system integrates a complete end-to-end pipeline, including object detection, confidence calibration, space counting, and real-time visualization through an interactive web-based dashboard. The system is designed to operate efficiently even in high-density parking scenarios, supporting large-scale detection with the capability to process hundreds to thousands of parking spaces simultaneously. Additionally, it incorporates techniques such as agnostic non-maximum suppression to eliminate overlapping detections and improve prediction reliability. A user-friendly Streamlit interface provides live analytics, including total parking spaces, available spaces, and occupancy percentage, enabling better decision-making for parking management. The key contributions of this work are as follows:

- Development of a real-time parking detection system using YOLOv8 for accurate classification of parking

spaces.

- Integration of a dynamic confidence calibration mechanism to improve prediction interpretability.
- Implementation of a scalable system capable of handling high-density parking environments.
- Design of an interactive dashboard for real-time monitoring and analytics.

The proposed system aims to enhance parking efficiency, reduce congestion, and contribute toward the development of intelligent transportation systems in smart cities.

II. Literature Review

The problem of parking space occupancy detection has been widely studied using vision-based techniques, particularly with the introduction of benchmark datasets such as PKLot. The PKLot dataset, introduced by Almeida et al., provides a large-scale collection of parking lot images captured under diverse environmental conditions, including varying illumination, weather, and camera viewpoints. It contains thousands of images with labeled parking spaces, making it highly suitable for developing robust parking detection systems [1]. Early approaches using the PKLot dataset relied on handcrafted features combined with classical machine learning techniques such as Support Vector Machines (SVM). Although these methods achieved reasonable accuracy in controlled environments, their performance degraded significantly under changing lighting conditions and different parking layouts, indicating poor generalization capability [1]. To overcome these limitations, deep learning-based methods were introduced. Amato et al. proposed a Convolutional Neural Network (CNN)-based approach for parking occupancy detection using smart cameras. Their method demonstrated improved robustness to environmental variations such as shadows and illumination changes. However, this approach required predefined parking slot regions, which limits scalability and adaptability to different camera angles and layouts [2]. Further advancements focused on automating the detection of parking slots. Grbić et al. proposed a system that combines vehicle detection with clustering techniques to automatically determine parking slot locations and classify occupancy. While this approach eliminates manual annotation, it introduces additional computational complexity and may affect real-time performance in practical applications [3]. Similarly, Nguyen et al. developed a smart camera-based parking system capable of detecting parking spaces and determining occupancy without manual intervention. Although the system improves automation, challenges such as occlusion, dense parking conditions, and varying viewpoints still affect its reliability [4]. Recent studies on modern deep learning-based detection models highlight that, despite achieving high accuracy, these systems still face challenges in real-world scenarios, including overlapping vehicles, small object detection, and inconsistent lighting conditions [5]. From the above literature, it is evident that while significant progress has been made in parking detection systems, several limitations remain. These include dependency on predefined parking slots, increased computational complexity in multi-stage systems, and reduced robustness in complex real-world environments. The proposed work addresses these challenges by developing a real-time smart parking system trained on the PKLot dataset. Unlike existing approaches, the system directly detects and classifies parking spaces without requiring predefined slot regions or complex intermediate processing. Additionally, it integrates real-time visualization and analytics, making it more scalable and suitable for practical deployment.

III. Dataset Description

The proposed system is developed and evaluated using the PKLot dataset, a widely recognized benchmark dataset for parking space detection and occupancy classification tasks. The dataset is publicly available through Roboflow and originates from the work of Almeida et al. The PKLot dataset consists of approximately 12,416 images captured from surveillance cameras installed at multiple parking locations. These images are collected under diverse environmental conditions, including sunny, cloudy, and rainy weather, enabling the development of models that are robust to real-world variations [1]. The dataset was acquired from three different parking areas: PUCPR, UFPR04, and UFPR05. Each location presents unique challenges such as varying ground textures, camera angles, and object occlusions.

Fig. 1. Sample image from PKLot dataset showing multiple parking spaces under normal lighting conditions

The images are typically of high resolution (approximately 1280×720 pixels), allowing detailed analysis of parking spaces. Each image contains multiple parking slots, usually ranging from 40 to 100 spaces. A key feature of the PKLot dataset is its large number of annotated parking spaces, with nearly 695,000 labeled instances categorized as either *occupied* or *empty* [1].

Fig. 2. Dataset variability showing different weather and lighting conditions (sunny, cloudy, rainy)

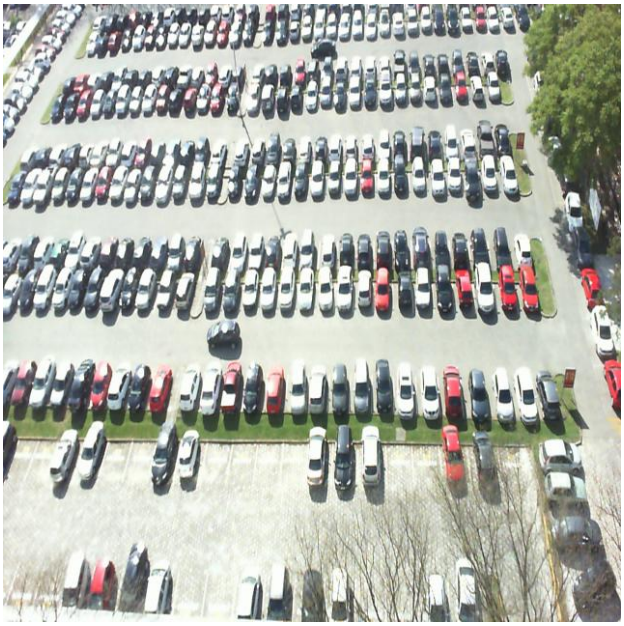

Fig. 3. Example of annotated parking spaces with bounding boxes for occupied and empty slots

The annotations in the dataset are provided as bounding boxes in the Roboflow version, making them directly compatible with object detection models such as YOLO. The dataset contains two primary classes:

- **space-empty**
- **space-occupied**

Additionally, the dataset captures temporal variations, as images are collected at regular intervals, enabling the model to learn real-world parking dynamics.

One of the major advantages of the PKLot dataset is its diversity and scale, which helps in training robust models capable of handling occlusion, overlapping vehicles, shadows, and varying lighting conditions. However, the dataset has limited night-time samples, which remains a challenge for real-world deployment. Overall, the PKLot dataset provides a comprehensive and realistic benchmark for developing vision-based smart parking systems.

IV. Proposed Methodology

The proposed system, ParkVision AI, is designed as an end-to-end vision-based smart parking detection framework that processes video input to determine parking space occupancy in real time. The system integrates deep learning-based object detection with post-processing and visualization modules to provide accurate and scalable parking analytics. The overall pipeline of the system is illustrated in Fig. 4. The methodology consists of five major stages: input acquisition, object detection, confidence calibration, space counting, and visualization.

A. Stage 1: Input Acquisition

The system accepts input in the form of images, pre-recorded videos, or live CCTV streams (RTSP/HTTP). This flexibility allows the system to be deployed in various real-world environments such as parking lots, campuses, and commercial areas. The input stream is processed frame-by-frame using OpenCV. Each frame is resized according to the selected inference resolution to balance detection accuracy and computational efficiency.

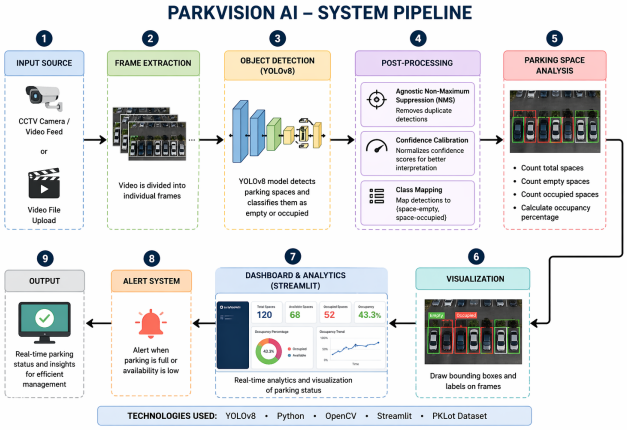

Fig. 4. Overall pipeline of the proposed smart parking system

B. Stage 2: Parking Space Detection using YOLOv8

Parking space detection is performed using the YOLOv8 object detection model, which is a single-stage deep learning architecture capable of performing localization and classification simultaneously. The model is trained on the PKLot dataset, where parking spaces are labeled into two classes:

- **space-empty**
- **space-occupied**

Given an input frame, the YOLOv8 model predicts bounding boxes along with class probabilities and confidence scores. The detection process can be mathematically expressed as:

$$D = f(I; \theta) \tag{1}$$

where I represents the input image, θ denotes the trained model parameters, and D is the set of detected bounding boxes with associated class labels and confidence scores. To handle dense parking scenarios, the system uses agnostic Non-Maximum Suppression (NMS), which eliminates overlapping detections regardless of class labels, ensuring that each parking space is counted only once.

C. Stage 3: Confidence Calibration

The raw confidence scores produced by the YOLOv8 model are often not directly interpretable due to variations in model output distribution. To address this, a normalization-based confidence calibration mechanism is applied. The calibrated confidence score is computed as:

$$C_{norm} = \min\left(0.99, \left(\frac{C_{raw}}{\alpha}\right)^{\beta}\right) \tag{2}$$

where C_{raw} is the original confidence score, and α and β are scaling parameters used to normalize the output distribution. This step improves the usability of the system by allowing users to set a meaningful confidence threshold through the interface.

D. Stage 4: Parking Space Counting and Analysis

After filtering detections based on the confidence threshold, the system categorizes each detected space as either occupied or empty. The total number of parking spaces is calculated as:

$$Total = N_{empty} + N_{occupied} \tag{3}$$

The parking availability percentage is computed as:

$$Availability = \frac{N_{empty}}{Total} \times 100 \tag{4}$$

Additionally, the system determines whether the parking area is fully occupied using a logical condition:

$$IsFull = \begin{cases} 1, & \text{if } N_{empty} = 0 \\ 0, & \text{otherwise} \end{cases} \tag{5}$$

These computed metrics form the basis of real-time parking analytics.

E. Stage 5: Visualization and User Interface

The final stage involves visualizing the detection results and analytics through an interactive Streamlit dashboard. Each detected parking space is represented by a bounding box:

- Green box for *space-empty*
- Red box for *space-occupied*

A dynamic dashboard overlay is generated on each frame, displaying:

- Total parking spaces
- Available spaces
- Occupied spaces
- Availability percentage

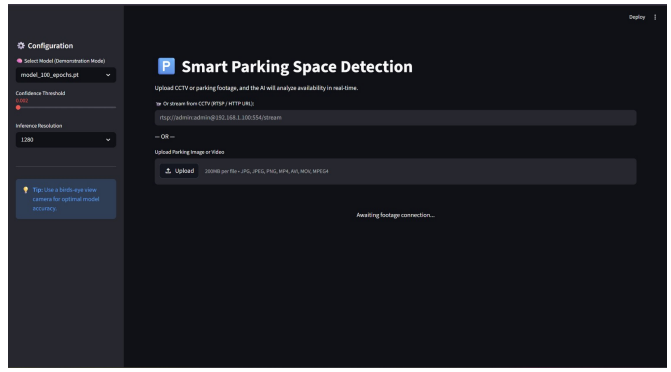

Fig. 5. Real-time detection output with parking analytics dashboard

The processed frames are continuously updated to provide real-time feedback to the user. In addition, an alert mechanism is implemented to notify when all parking spaces are occupied.

F. System Advantages

The proposed methodology offers several advantages:

- Real-time performance using a single-stage detection model
- Scalability to high-density parking environments
- No dependency on predefined parking slot mapping
- Cost-effective deployment using only camera-based input

Overall, the system provides a robust and scalable solution for intelligent parking management using deep learning and computer vision techniques.

V. Model Architecture and Implementation

A. YOLOv8 Model Architecture

The proposed system utilizes the YOLOv8 (You Only Look Once version 8) object detection model for identifying and classifying parking spaces. YOLOv8 is a single-stage detection framework that performs both localization and classification in a unified architecture, making it highly suitable for real-time applications. The architecture of YOLOv8 consists of three primary components: backbone, neck, and detection head. The backbone is responsible for feature extraction and is based on a convolutional neural network that captures spatial and semantic information from the input image. The neck employs feature aggregation techniques, such as Path Aggregation Network (PAN), to combine multi-scale features and improve detection performance across objects of varying sizes. The detection head is anchor-free and directly predicts bounding box coordinates along with class probabilities, eliminating the need for predefined anchor boxes. The model processes an input image of resolution $N \times N$ and generates a set of predictions in the form:

$$P = \{(x_i, y_i, w_i, h_i, c_i, p_i)\}_{i=1}^{n} \tag{6}$$

where (x_i, y_i) represents the center coordinates of the bounding box, (w_i, h_i) denote its width and height, c_i is the confidence score, and p_i is the predicted class label. The use of an anchor-free detection head simplifies the training process and improves generalization, particularly for datasets such as PKLot where parking spaces vary in size and orientation.

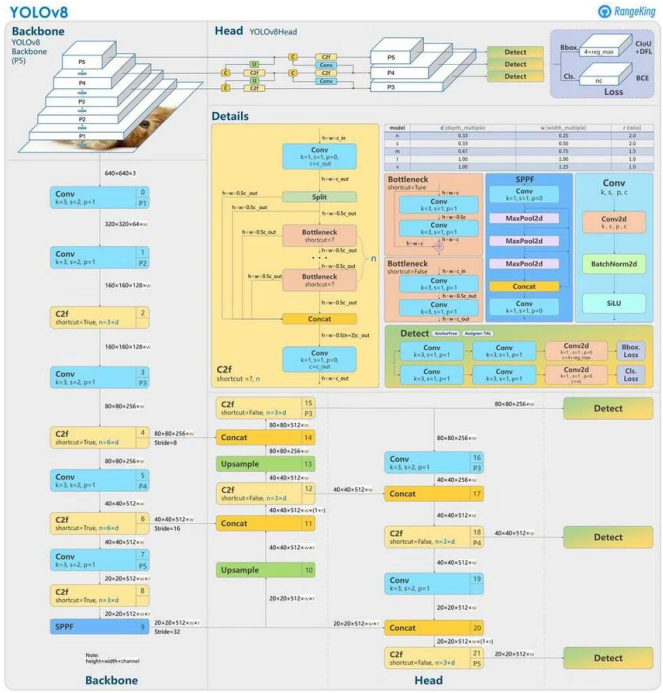

Fig. 6. YOLOv8 architecture showing backbone, neck, and detection head

B. Model Training and Weights

The model is trained using the PKLot dataset, where each parking space is annotated with bounding boxes and labeled as either *space-empty* or *space-occupied*. The training process involves optimizing the model parameters to minimize localization and classification loss. The final trained weights are stored in a file (best.pt), which represents the model with the best validation performance. During inference, the system dynamically loads the most recent or available weight file, ensuring flexibility and ease of deployment.

C. Implementation Framework

The system is implemented using Python and integrates multiple libraries and frameworks for efficient execution. The core components of the implementation include:

- **Ultralytics YOLOv8:** Used for model training and inference.
- **OpenCV:** Handles video processing, frame extraction, and visualization.
- **Streamlit:** Provides an interactive web-based user interface.
- **NumPy and Pandas:** Used for numerical computations and data handling.

The application follows a modular design, where the detection logic is encapsulated within a dedicated class responsible for training, inference, and evaluation. This modular approach improves code maintainability and scalability.

D. Inference Pipeline

During runtime, the system processes input frames sequentially and performs inference using the trained YOLOv8 model. The key steps in the inference pipeline are as follows:

1) Capture frame from video stream or image input.
2) Perform object detection using YOLOv8.
3) Apply agnostic Non-Maximum Suppression (NMS) to remove redundant detections.
4) Normalize confidence scores using calibration.
5) Classify detected spaces as empty or occupied.
6) Compute parking statistics and visualize results.

The inference process is optimized for real-time performance by adjusting parameters such as image resolution and confidence thresholds.

E. User Interface and Deployment

The system includes a Streamlit-based graphical user interface that enables users to upload videos, stream live camera feeds, and visualize detection results in real time. The interface provides configurable parameters such as confidence threshold and image resolution, allowing users to control the performance of the system. The dashboard dynamically updates key metrics, including total parking spaces, available spaces, and occupancy status. Additionally, visual overlays are rendered on the processed frames to enhance interpretability. The system can be deployed on local machines or cloud environments and supports both CPU and GPU-based execution, making it adaptable to different hardware configurations.

F. System Efficiency

The use of a single-stage detection model ensures low latency and high throughput, enabling the system to process multiple frames per second. The lightweight architecture of YOLOv8 further enhances computational efficiency, making the solution suitable for real-time applications without requiring specialized hardware. Overall, the integration of YOLOv8 with an optimized inference pipeline and an interactive dashboard results in a robust and scalable smart parking detection system.

VI. Experiments and Results

A. Detection Performance

The performance of the proposed model is evaluated using standard object detection metrics, including Precision, Recall, and mean Average Precision (mAP). These metrics provide a comprehensive measure of the model's ability to correctly detect and classify parking spaces.

Precision represents the proportion of correctly identified parking spaces among all detections, while Recall indicates the model's ability to detect all actual parking spaces. The

TABLE I
MODEL PERFORMANCE

Model	Precision	Recall	mAP
YOLOv8	0.953	0.788	0.876

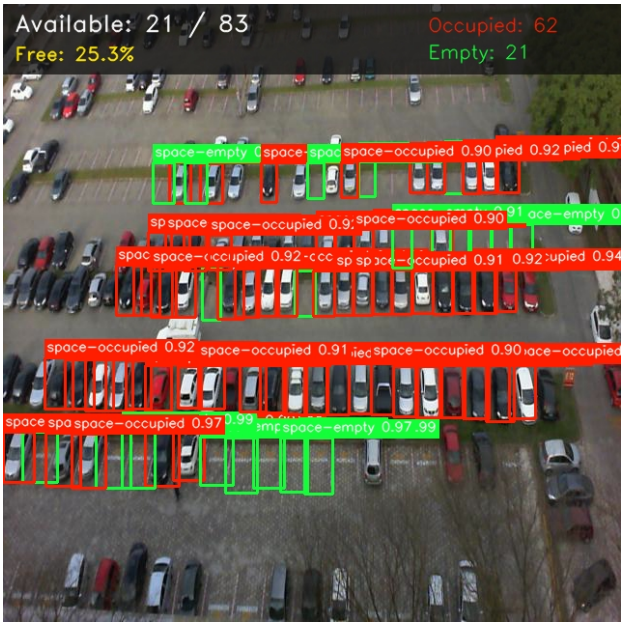

Fig. 7. Sample detection results showing classification of parking spaces as occupied and empty

mAP metric summarizes the overall detection accuracy across different confidence thresholds.

The results demonstrate that the YOLOv8 model is capable of accurately detecting parking spaces even in dense and complex scenarios. The model performs well under varying lighting conditions and maintains consistent detection performance across different parking layouts.

B. System Output

The developed system provides real-time parking analytics through a visual dashboard. The key outputs of the system include:

- Total number of parking spaces detected
- Number of available (empty) spaces
- Number of occupied spaces
- Real-time occupancy percentage
- Visual bounding boxes for each detected parking space

The system successfully processes both images and video streams, providing continuous updates of parking availability. Additionally, an alert mechanism is triggered when all parking spaces are occupied, enhancing usability in real-world scenarios.

VII. CONCLUSION AND FUTURE SCOPE

A. Conclusion

- This paper presented ParkVision AI, a real-time smart parking detection system based on YOLOv8 and computer vision techniques.
- The system successfully detects and classifies parking spaces as occupied or empty from images and video streams.
- A vision-based approach eliminates the need for additional hardware, making the solution cost-effective and scalable.
- The integration of confidence calibration and efficient post-processing improves detection reliability in complex scenarios.
- The system provides real-time analytics through an interactive dashboard, enabling effective parking management.

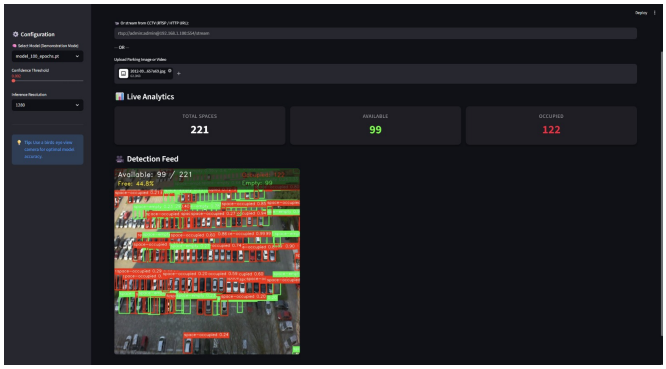

Fig. 8. Real-time dashboard displaying parking statistics and detection results

B. Future Scope

- Integration of multi-camera support for monitoring large-scale parking areas.
- Development of a mobile application for improved accessibility and user interaction.
- Incorporation of IoT-based automation for smart parking infrastructure.
- Enhancement of detection performance under low-light and night-time conditions.

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